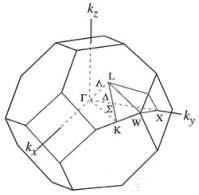
Bipolar Junction Transistors



Current Gain

Emitter injection efficiency

$$\gamma \approx \frac{1}{1 + \frac{N_B}{N_E} \cdot \frac{D_E}{D_B} \cdot \frac{x_B}{x_E}} \qquad (x_B \ll L_B),\ (x_E \ll L_E)$$

Base transport factor

$$\alpha_T \approx \frac{1}{1 + \frac{1}{2}\left(\frac{x_B}{L_B}\right)^2} \qquad (x_B \ll L_B)$$

Recombination factor

$$\delta = \frac{1}{1 + \frac{J_{r0}}{J_{s0}} \exp\left(\frac{-eV_{BE}}{2kT}\right)}$$

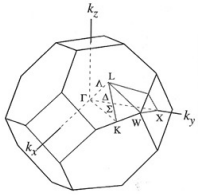
Common-base current gain

$$\alpha = \gamma \alpha_T \delta \approx \frac{1}{1 + \frac{N_B}{N_E} \cdot \frac{D_E}{D_B} \cdot \frac{x_B}{x_E} + \frac{1}{2}\left(\frac{x_B}{L_B}\right)^2 + \frac{J_{r0}}{J_{s0}} \exp\left(\frac{-eV_{BE}}{2kT}\right)}$$

Common-emitter current gain

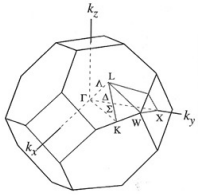
$$\beta = \frac{\alpha}{1 - \alpha} \approx \frac{1}{\frac{N_B}{N_E} \cdot \frac{D_E}{D_B} \cdot \frac{x_B}{x_E} + \frac{1}{2}\left(\frac{x_B}{L_B}\right)^2 + \frac{J_{r0}}{J_{s0}} \exp\left(\frac{-eV_{BE}}{2kT}\right)}$$

$$\beta_{dc} \approx \frac{D_n}{W_B} \frac{W_E}{D_p} \frac{n_{i,B}^2}{n_{i,E}^2} \frac{N_E}{N_B}$$



Nonideal Effect

- Base width modulation
- Current Crowding
- Punch-Through
- High Injection
- Emitter Bandgap Narrowing



Problem of Low Base Doping: Base Width Modulation

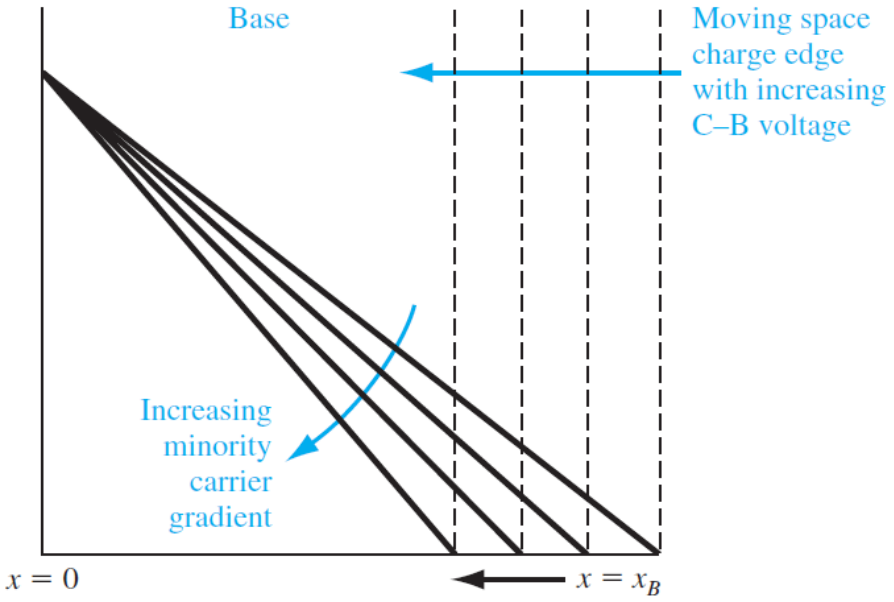
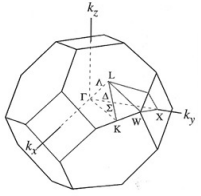
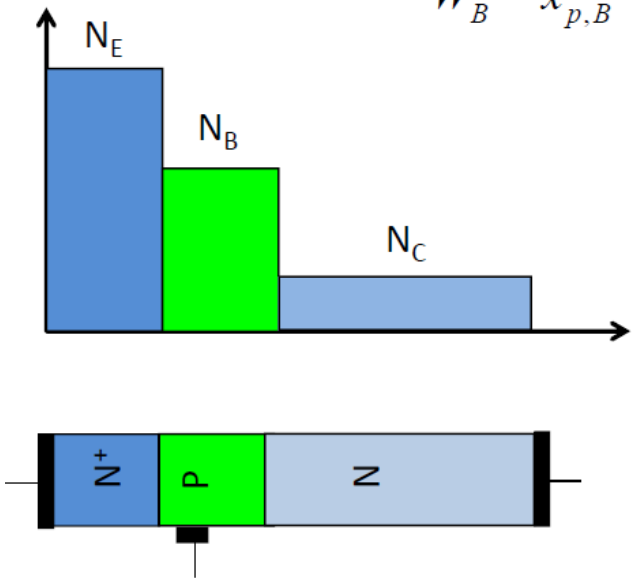


Figure 12.21 | The change in the base width and the change in the minority carrier gradient as the B–C space charge width changes.



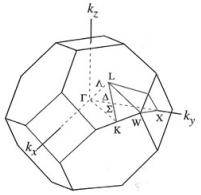
Problem of Low Base Doping: Base Width Modulation

$$\beta_{DC} \approx \frac{D_n}{W_B - x_{p,B} - x_{p,c}} \frac{W_E}{D_p} \frac{n_{i,B}^2}{n_{i,E}^2} \frac{N_E}{N_B}$$



$$x_{p,BE} = \sqrt{\frac{2k_s\varepsilon_0}{q} \frac{N_E}{N_B(N_E + N_B)}(V_{bi} - V_{BE})}$$

$$x_{p,BC} = \sqrt{\frac{2k_s\varepsilon_0}{q} \frac{N_C}{N_B(N_C + N_B)}(V_{bi} - V_{BC})}$$



Gain depends on collector voltage (bad) …

Problem of Low Base Doping: Early Effect

$$|J_C| = eD_B \frac{d[\delta n_B(x)]}{dx} \cong \frac{eD_B n_{B0}}{x_B} \exp\left(\frac{V_{BE}}{V_t}\right)$$

$$\frac{dI_C}{dV_{CE}} \equiv g_o = \frac{I_C}{V_{CE} + V_A} = \frac{1}{r_o}$$

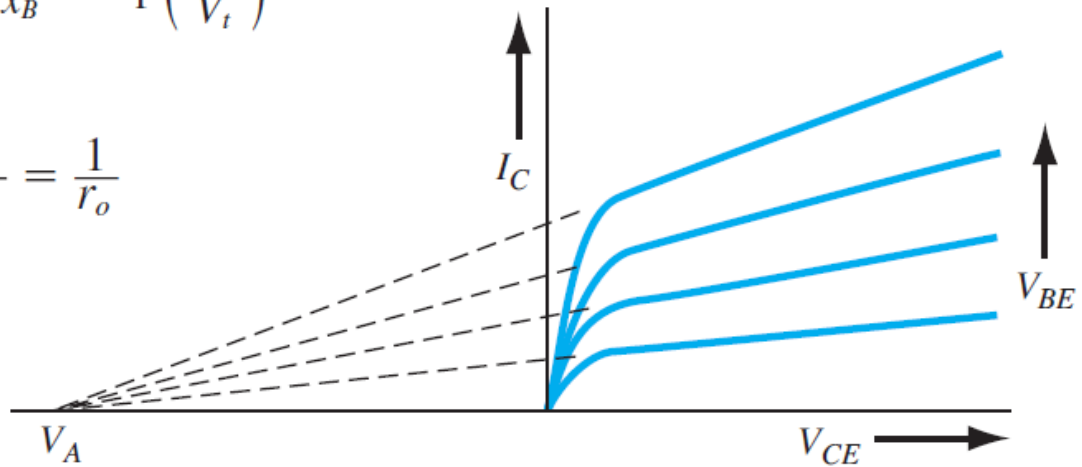
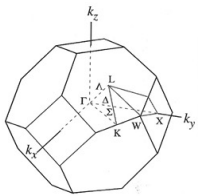


Figure 12.22 | The collector current versus collector–emitter voltage showing the Early effect and Early voltage.



Problem of Low Base Doping: Current Crowding

- When the base doping is low, there is an internal base resistance under the emitter
- At high current, the resistive voltage drop reduces the net V_{BE} across the junction

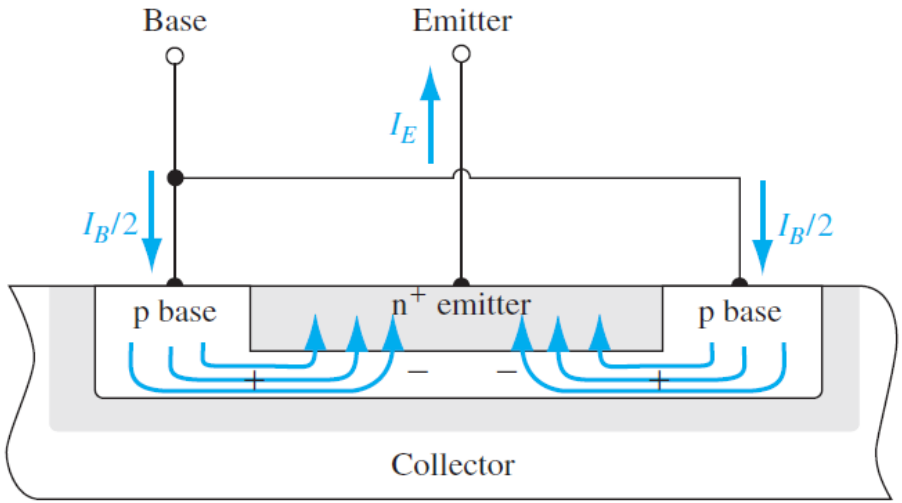
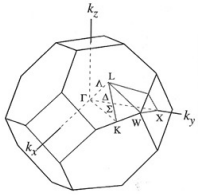
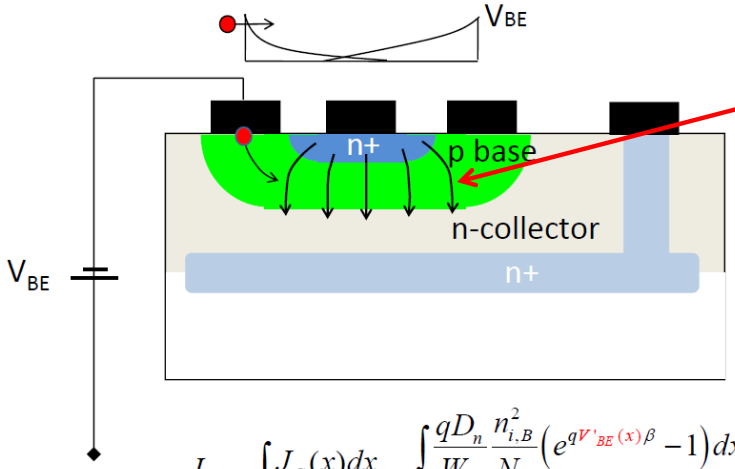


Figure 12.27 | Cross section of an npn bipolar transistor showing the base current distribution and the lateral potential drop in the base region.



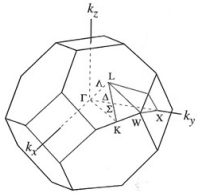
Problem of Low Base Doping: Current Crowding

- When the base doping is low, there is an internal base resistance under the emitter
- At high current, the resistive voltage drop reduces the net V_{BE} across the junction



More current at the edge causes localized heating

$$\beta = \frac{I_C}{I_B} = \frac{\int J_C(x)dx}{\int J_B(x)dx} = \frac{\int \frac{qD_n}{W_B}\frac{n_{i,B}^2}{N_B}\left(e^{qV'_{BE}(x)\beta}-1\right)dx}{\int \frac{qD_p}{W_E}\frac{n_{i,E}^2}{N_E}\left(e^{qV'_{BE}(x)\beta}-1\right)dx}$$



Interdigitated comb-like Contact

To increase periphery to area ratio.

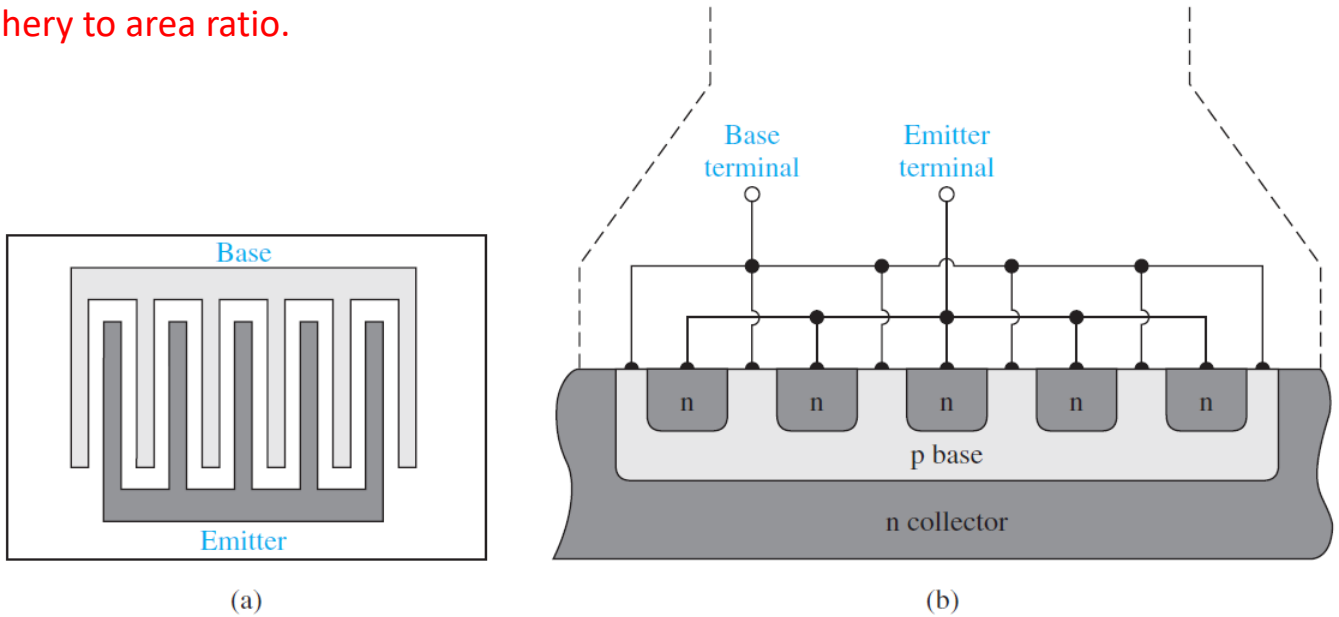
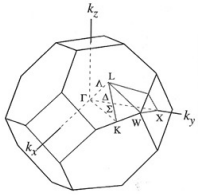
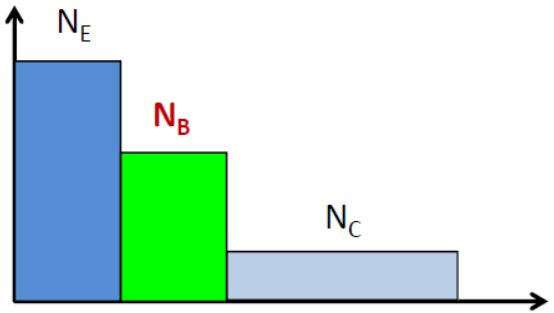
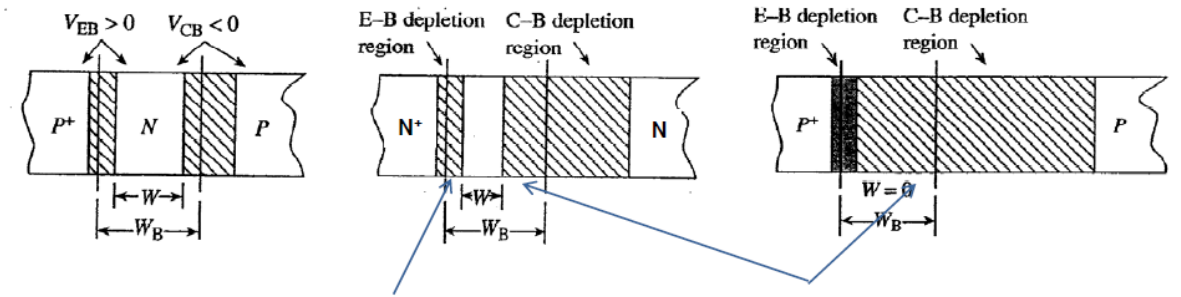


Figure 12.29 | (a) Top view and (b) cross section of an interdigitated npn bipolar transistor structure.



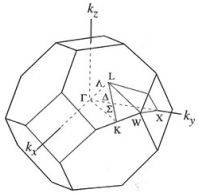
Problem of Low Base Doping: Punch Through





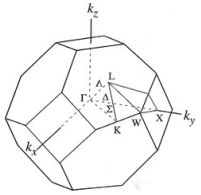
$$x_{p,BE} = \sqrt{\frac{2k_s\varepsilon_0}{q}\frac{N_E}{N_B(N_E+N_B)}(V_{bi}-V_{BE})}$$

$$x_{p,BC} = \sqrt{\frac{2k_s\varepsilon_0}{q}\frac{N_C}{N_B(N_C+N_B)}(V_{bi}-V_{BC})}$$

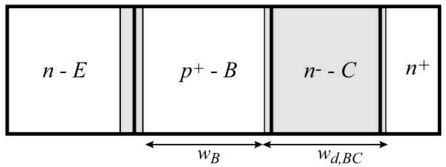


Collector Doping

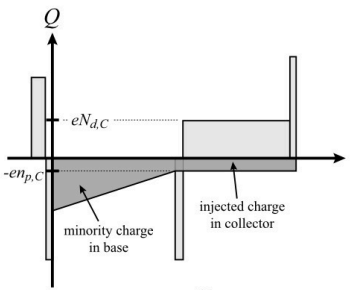
- How can you solve the early voltage issue and punch-through issue?
- Reduce the collector doping? $N_C < N_B$



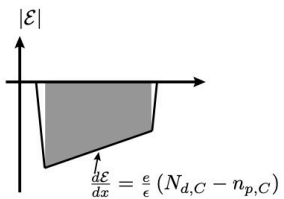
Kirk Effect and Base Pushout



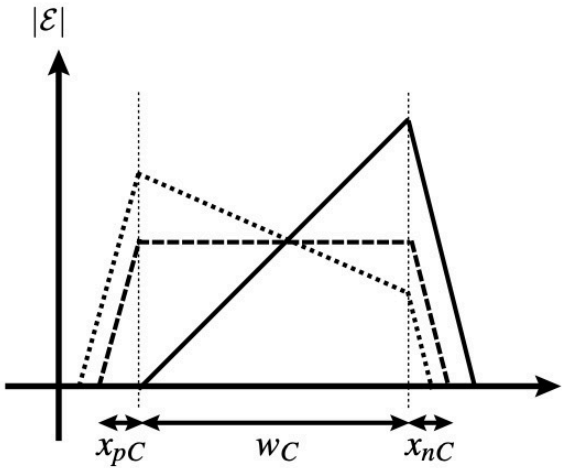
(a)



(b)



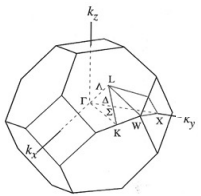
(c)



$J_C < J_{crit}$

$J_C = J_{crit}$

$J_C = J_{Kirk}$



High Injection due to Generation-Recombination

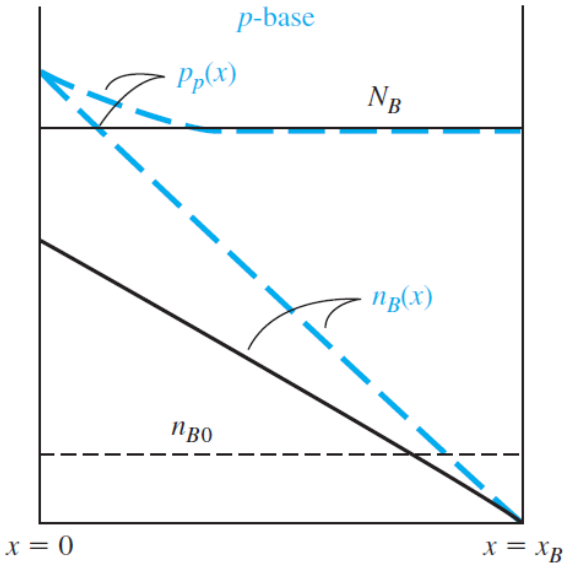
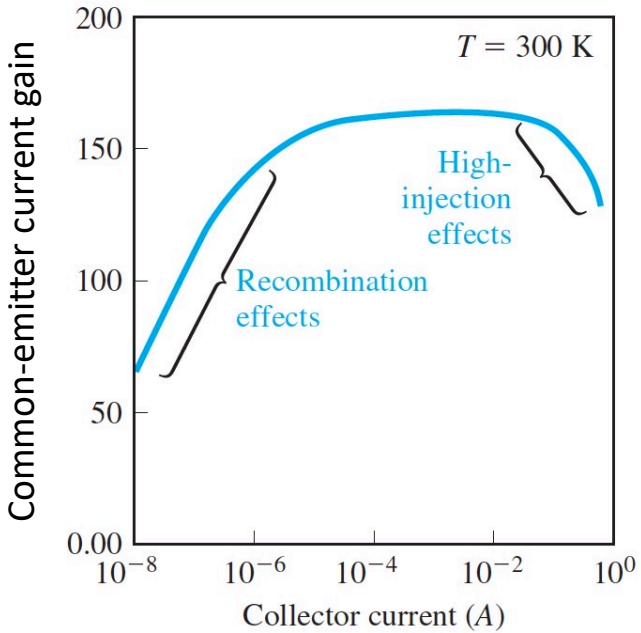
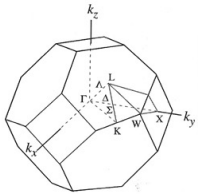


Figure 12.23 | Minority and majority carrier concentrations in the base under low and high injection (solid line: low injection; dashed line: high injection).





Perhaps High Doping in Emitter?

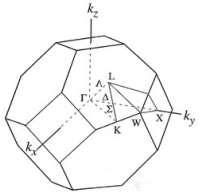
- Unfortunately…

$$n_i^2 = N_c N_v \exp\left(\frac{-E_g}{kT}\right)$$

$$n_{iE}^2 = N_c N_v \exp\left[\frac{-(E_{g0} - \Delta E_g)}{kT}\right] = n_i^2 \exp\left(\frac{\Delta E_g}{kT}\right)$$

$$\gamma = \frac{1}{1 + \dfrac{p_{E0} D_E L_B}{n_{B0} D_B L_E} \cdot \dfrac{\tanh\,(x_B/L_B)}{\tanh\,(x_E/L_E)}}$$

$$p'_{E0} = \frac{n_{iE}^2}{N_E} = \frac{n_i^2}{N_E} \exp\left(\frac{\Delta E_g}{kT}\right)$$



Breakdown Voltage

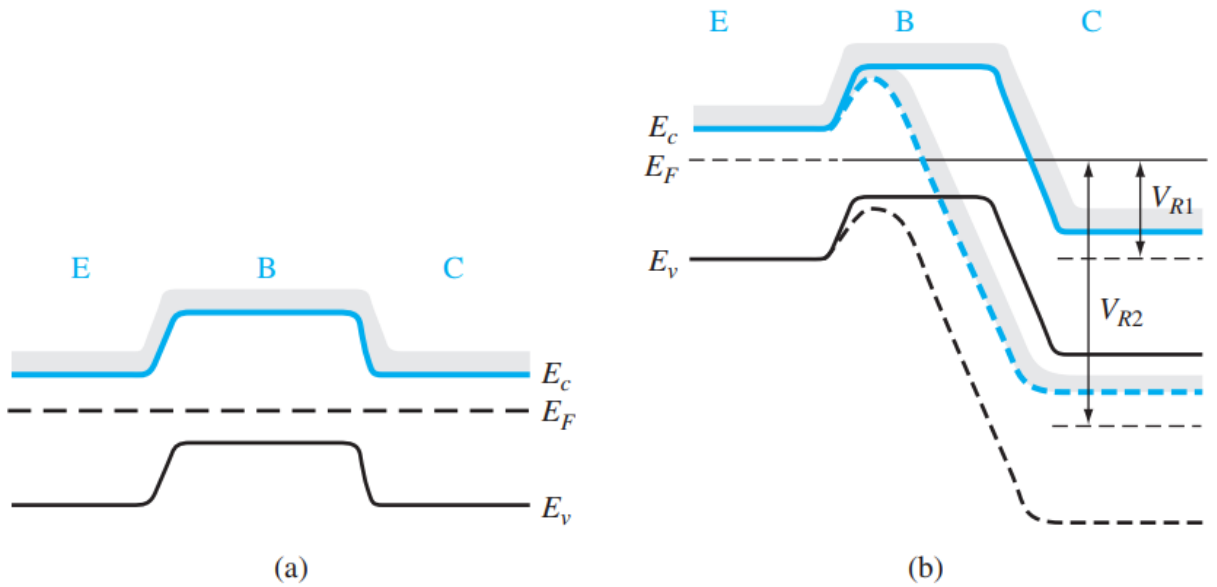
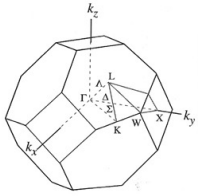


Figure 12.32 | Energy-band diagram of an npn bipolar transistor (a) in thermal equilibrium, and (b) with a reverse-biased B–C voltage before punch-through, V_{R1}, and after punch-through, V_{R2}.



Breakdown Voltage

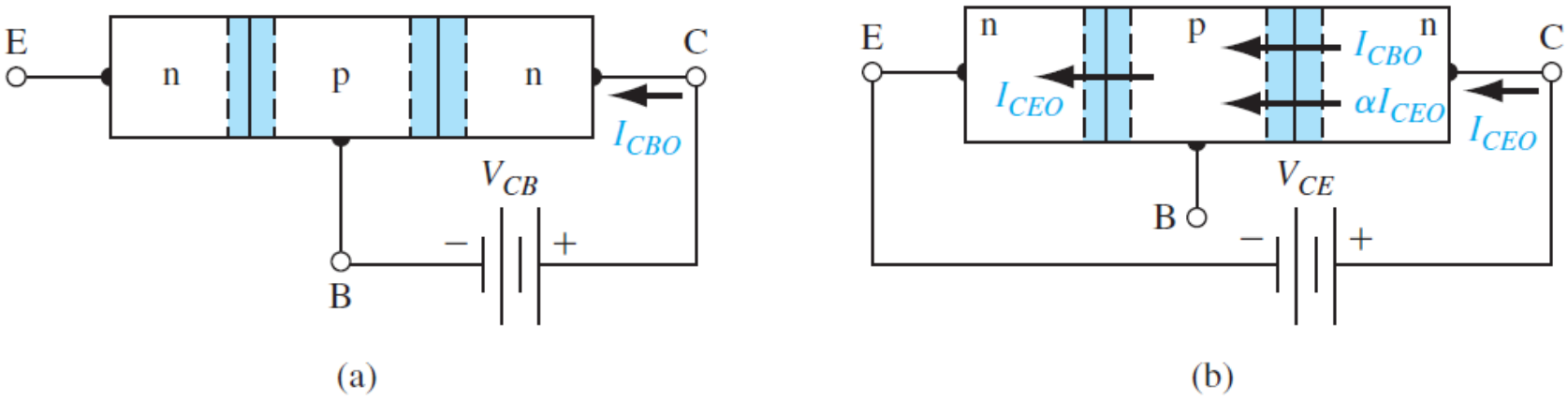
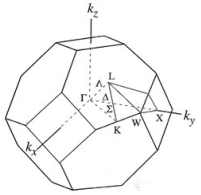


Figure 12.34 | (a) Open-emitter configuration with saturation current I_{CBO}: (b) Open-base configuration with saturation current I_{CEO}.



Breakdown Voltage

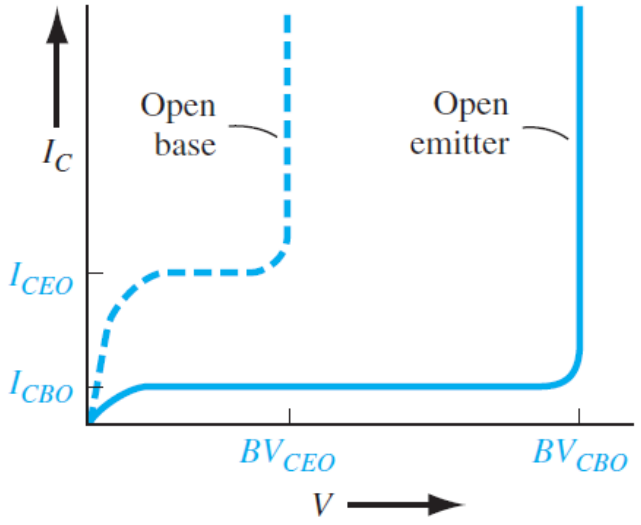
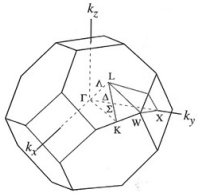


Figure 12.35 | Relative breakdown voltages and saturation currents of the open-base and open-emitter configurations.



Hybrid-Pi Model

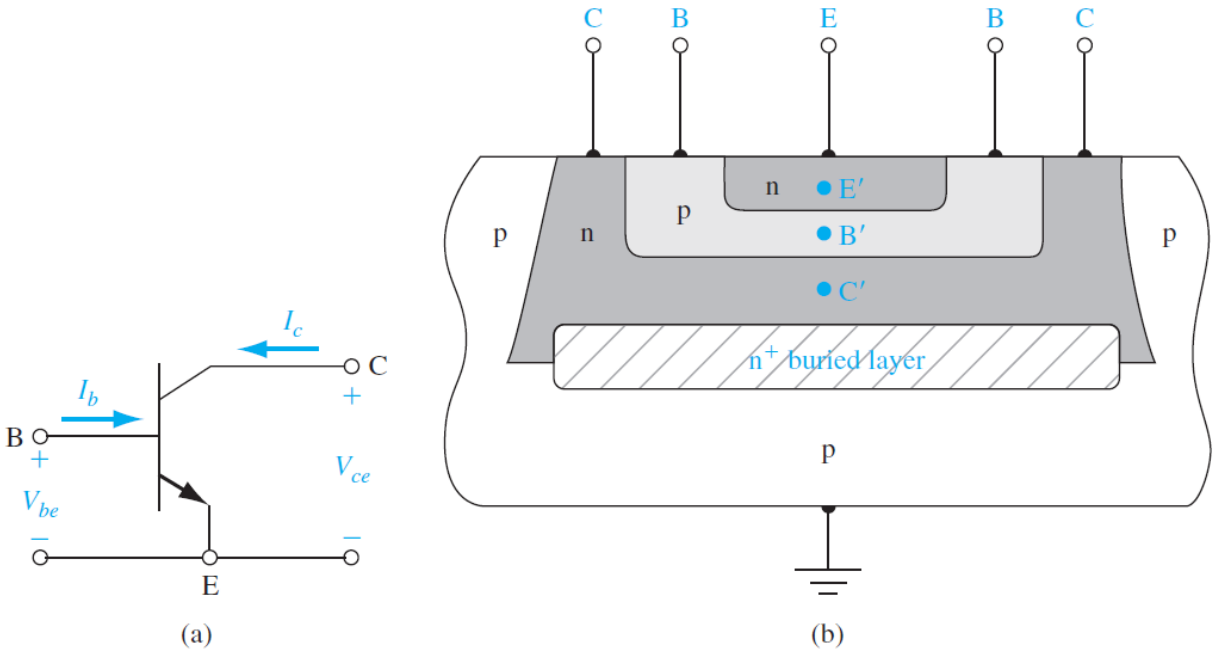
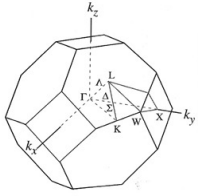


Figure 12.38 | (a) Common-emitter npn bipolar transistor with small-signal current and voltages. (b) Cross section of an npn bipolar transistor for the hybrid-pi model.



Hybrid-Pi Model

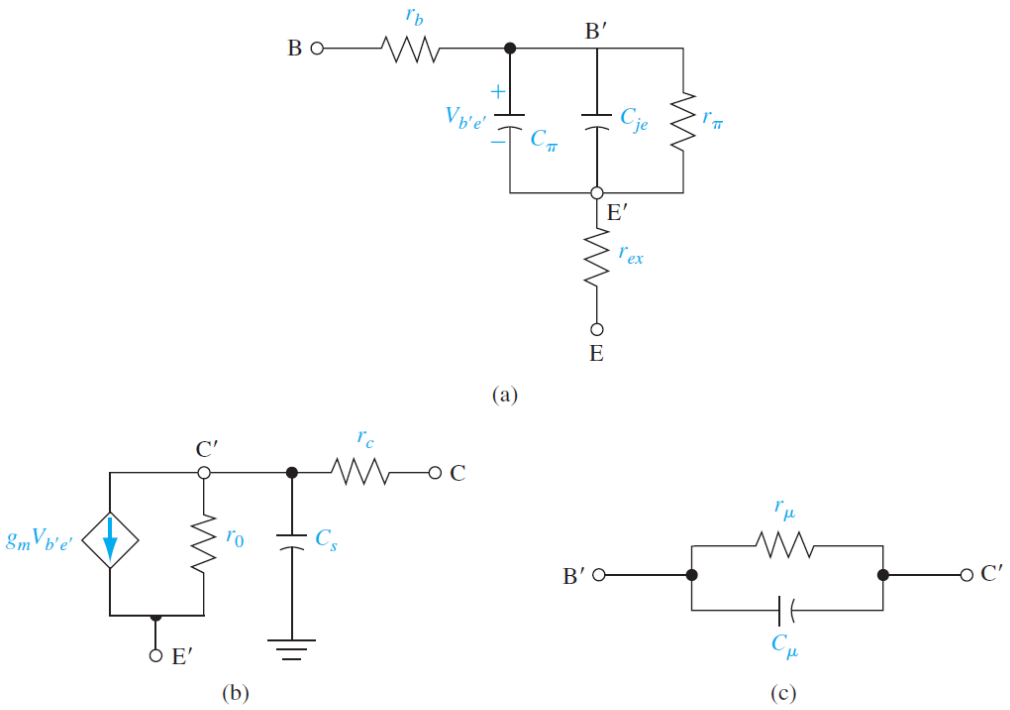
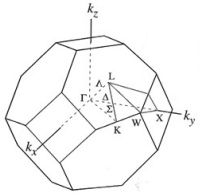


Figure 12.39 | Components of the hybrid-pi equivalent circuit between (a) the base and emitter, (b) the collector and emitter, and (c) the base and collector.



Hybrid-Pi Equivalent Circuit

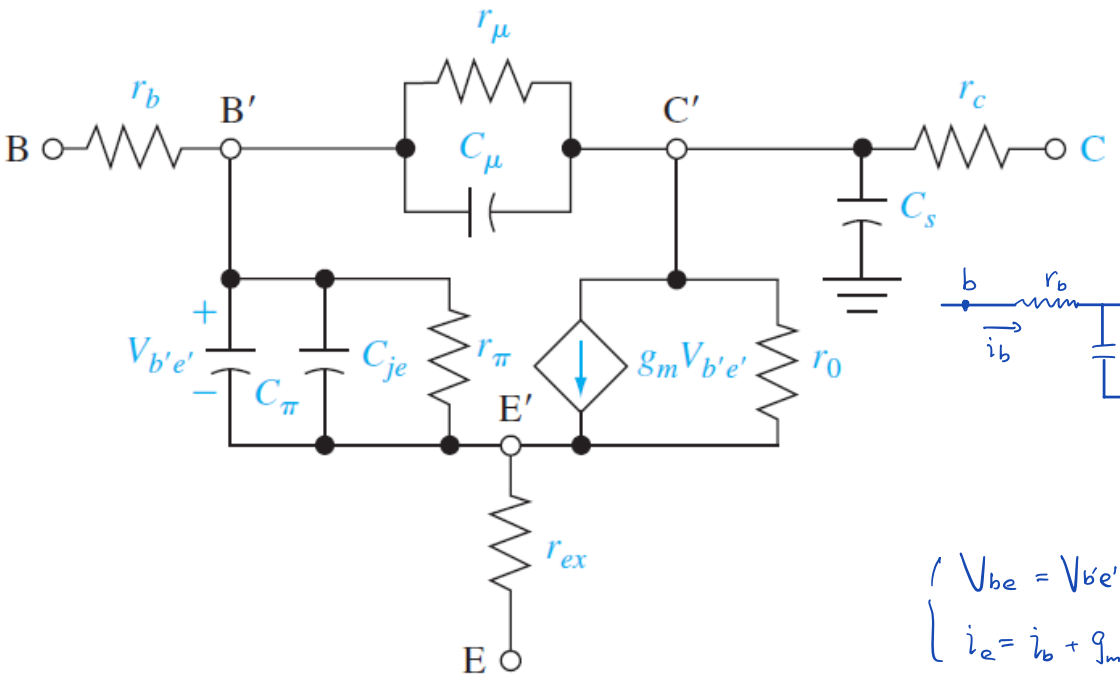
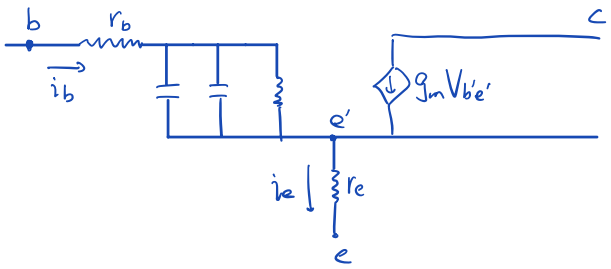


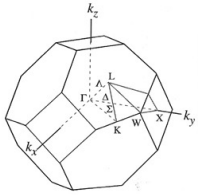
Figure 12.40 | Hybrid-pi equivalent circuit.



$$\begin{cases} V_{be} = V_{b'e'} + i_b r_b + i_e r_e \\ i_e = i_b + g_m V_{b'e'} \end{cases}$$

$$V_{be} = V_{b'e'} + i_b r_b + (i_b + g_m V_{b'e'}) r_e \cong (1 + g_m r_e) V_{b'e'}$$

$$i_c = g_m' V_{be} = g_m V_{b'e'} \quad \Rightarrow g_m' = g_m \frac{V_{b'e'}}{V_{be}} = \frac{g_m}{1 + g_m r_e}$$



Example

Objective: Determine, to a first approximation, the frequency at which the small-signal current gain decreases to $1/\sqrt{2}$ of its low-frequency value.

Consider the simplified hybrid-pi circuit shown in Figure 12.41. We are ignoring C_μ, C_s, r_μ, C_{je}, r_0, and the series resistances. We must emphasize that this is a first-order calculation and that C_μ normally cannot be neglected.

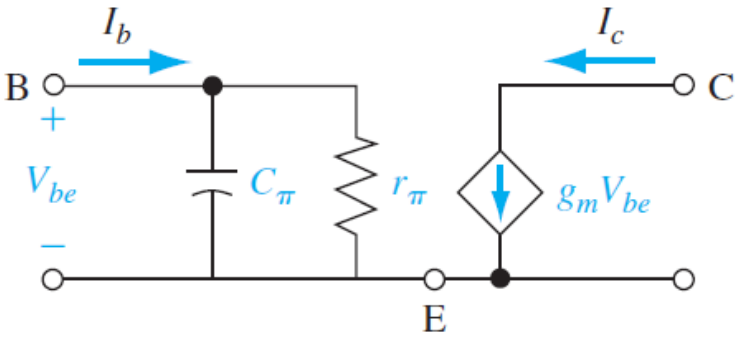
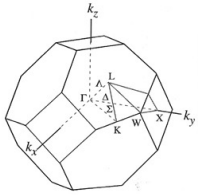


Figure 12.41 | Simplified hybrid-pi equivalent circuit.

$$V_{be} = \frac{\frac{r_\pi}{j\omega C_\pi} I_b}{r_\pi + \frac{1}{j\omega C_\pi}} = \frac{r_\pi}{1 + j\omega C_\pi r_\pi} \cdot I_b$$

$$I_c = \frac{g_m r_\pi}{1 + j\omega C_\pi r_\pi} \cdot I_b$$

$$\Rightarrow \beta = \frac{g_m r_\pi}{1 + j\omega C_\pi r_\pi}$$

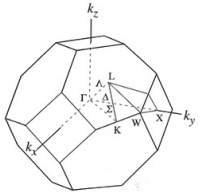


Frequency Limitations

- Time Delay Factors

$$\tau_{ec} = \tau_e + \tau_b + \tau_d + \tau_c$$

τ_{ec} = emitter-to-collector time delay
τ_e = emitter–base junction capacitance charging time
τ_b = base transit time
τ_d = collector depletion region transit time
τ_c = collector capacitance charging time



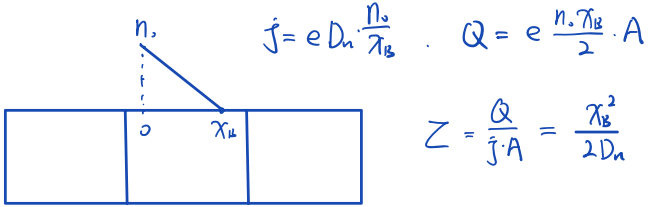
Frequency Limitations

- Emitter-Base Junction Capacitance Charging Time

$$\tau_e = r'_e\,(C_{je} + C_p) \qquad r'_e = \frac{kT}{e}\cdot\frac{1}{I_E}$$

- Base Transit Time:

$$\tau_b = \frac{x_B^2}{2D_n}$$

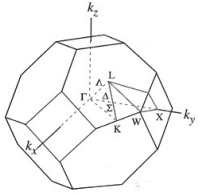


- Collector Depletion Region Transit Time:

$$\tau_d = \frac{x_{dc}}{v_s}$$

- Collector Capacitance Charging Time

$$\tau_c = r_c(C_\mu + C_s)$$

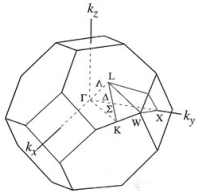


Transistor Cutoff Frequency

$$\alpha = \frac{\alpha_0}{1 + j\dfrac{f}{f_\alpha}} \qquad f_\alpha = \frac{1}{2\pi\tau_{ec}} \qquad \beta = \frac{\alpha}{1-\alpha}$$

$$\beta = \frac{\alpha}{1-\alpha} = \frac{\dfrac{\alpha_0}{1 + j(f/f_T)}}{1 - \dfrac{\alpha_0}{1 + j(f/f_T)}} = \frac{\alpha_0}{1 - \alpha_0 + j(f/f_T)}$$

$$\beta = \frac{\alpha_0}{(1-\alpha_0)\left[1 + j\dfrac{f}{(1-\alpha_0)f_T}\right]} \approx \frac{\beta_0}{1 + j\dfrac{\beta_0 f}{f_T}}$$



Bode Plot

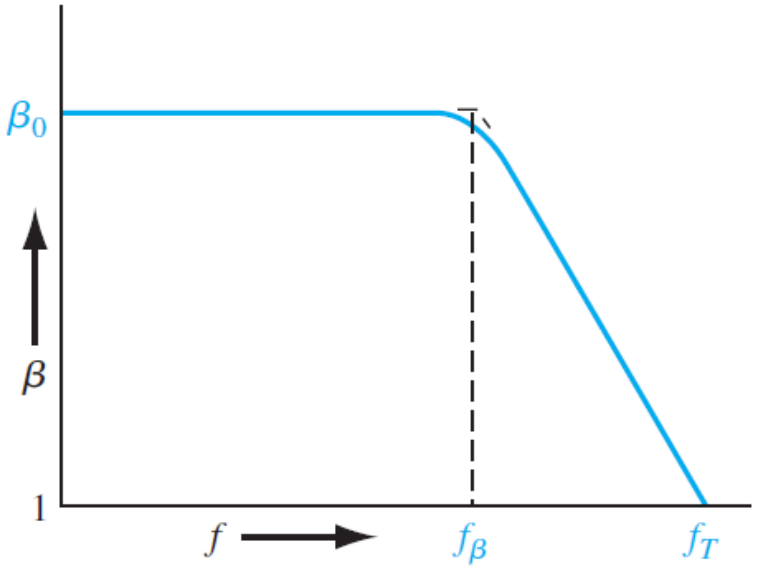


Figure 12.42 | Bode plot of common-emitter current gain versus frequency.

